



TECHNOLOGY INNOVATION HUB – TIDF INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

ACADEMIC INTERNSHIP REPORT

Brain Tumor MRI Segmentation

A Deep Learning-Based Approach for Automated Brain Tumor Segmentation

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TIH-TIDF Summer Internship Program

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ABSTRACT

Brain tumors are among the most serious neurological disorders, and accurate identification of tumor regions from Magnetic Resonance Imaging (MRI) scans is essential for diagnosis, treatment planning, and patient monitoring. Manual analysis of MRI images is time-consuming and may vary between radiologists, highlighting the need for automated and reliable segmentation techniques. This project presents a deep learning-based framework for the detection and segmentation of **Lower-Grade Gliomas (LGG)** using the publicly available LGG MRI Segmentation dataset.

The proposed system follows a two-stage approach. Initially, an ensemble of deep learning classification models, including **ResNet34, EfficientNet-B0, and DenseNet121**, is used to determine the presence of a tumor in an MRI image. If a tumor is detected, an ensemble of **U-Net-based segmentation models** with different encoder architectures generates a pixel-wise tumor segmentation mask. The final segmentation is obtained by combining the predictions of the individual models to improve robustness and accuracy.

To improve the interpretability of the system, several Explainable Artificial Intelligence (XAI) techniques, including **Gradient-weighted Class Activation Mapping (Grad-CAM), agreement maps**, and **Monte Carlo Dropout-based uncertainty estimation**, are incorporated to visualize the model's decision-making process and estimate prediction confidence.

The complete framework was developed and trained on a local workstation equipped with a **4 GB VRAM GPU**. By utilizing techniques such as **Automatic Mixed Precision (AMP), gradient accumulation**, and sequential model training, the system achieved a **classification accuracy of 95.70%** and a **Dice Score of 88.33%** on unseen patient data. Finally, the trained models were integrated into a **Flask-based web application**, providing an interactive interface for automated brain tumor analysis and demonstrating the potential of deep learning as a decision-support tool in medical imaging.

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Chapter 1: INTRODUCTION

1.1 Background

Brain tumors are abnormal growths of cells within the brain that can significantly affect a person's neurological functions and overall health. Among these, **Lower-Grade Gliomas (LGGs)** are slow-growing brain tumors that originate from glial cells. Although they grow more slowly than high-grade gliomas, they can progress over time and become more aggressive if not diagnosed and treated appropriately.

Magnetic Resonance Imaging (MRI) is the most commonly used imaging technique for detecting and monitoring brain tumors because it provides detailed images of soft tissues without exposing patients to ionizing radiation. During diagnosis, radiologists examine MRI scans to identify the location and size of tumors. However, manually outlining tumor boundaries in every MRI slice is a time-consuming and challenging task that may also lead to variations between different medical experts.

1.2 Problem Statement

Accurate segmentation of brain tumors from MRI images is essential for diagnosis, treatment planning, and monitoring disease progression. Manual segmentation requires considerable time and expertise and is prone to human variability. Therefore, there is a need for an automated system that can accurately detect and segment brain tumors while providing reliable and interpretable results for medical professionals.

1.3 Objectives of the Project

The main objectives of this project are:

- To develop an automated deep learning framework for brain tumor detection and segmentation.
 - To improve segmentation accuracy using an ensemble of deep learning models.
 - To incorporate Explainable Artificial Intelligence (XAI) techniques such as Grad-CAM and uncertainty estimation to improve model interpretability.
 - To evaluate the proposed system using standard performance metrics such as Dice Score and classification accuracy.
 - To develop a Flask-based web application for easy interaction with the trained models.
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1.4 Scope of the Project

This project focuses on the segmentation of **Lower-Grade Glioma (LGG)** tumors from MRI images using deep learning techniques. The system performs tumor classification followed by pixel-level segmentation and provides visual explanations of the model's predictions. The proposed framework is designed as a decision-support tool to assist healthcare professionals and demonstrates the practical application of artificial intelligence in medical image analysis.

Chapter 2: Literature Review

2.1 Brain Tumor Segmentation

Brain tumor segmentation is an important task in medical image analysis as it helps identify the exact location and boundaries of tumors from MRI scans. Traditionally, segmentation was performed manually by radiologists, which is time-consuming and subject to differences between observers. Earlier automated approaches used image processing techniques such as thresholding, region growing, and clustering algorithms. Although these methods were simple and computationally efficient, they often struggled with variations in tumor shape, size, and image intensity.

The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), significantly improved the performance of medical image segmentation by automatically learning important features directly from images.

2.2 Deep Learning in Medical Image Analysis

Deep learning has become one of the most effective approaches for medical image analysis due to its ability to learn complex patterns from large datasets. CNN-based models have been widely used for disease detection, image classification, and image segmentation.

Among these models, **U-Net** has become one of the most popular architectures for biomedical image segmentation. Its encoder-decoder structure with skip connections enables accurate localization while preserving fine image details. Several improved versions of U-Net have also been developed by incorporating powerful backbone networks such as **ResNet**, **EfficientNet**, and **DenseNet**, resulting in improved feature extraction and segmentation performance.

2.3 Existing Segmentation Techniques

Many recent studies have focused on improving segmentation accuracy using deep learning. While single-model architectures often produce good results, they may not generalize well on limited datasets and can be sensitive to variations in medical images. Ensemble learning has therefore emerged as an effective solution by combining predictions from multiple models to improve robustness and reduce prediction errors.

However, implementing ensemble models requires greater computational resources and memory, making them difficult to train on consumer-grade hardware.

Chapter 3: Dataset and Preprocessing

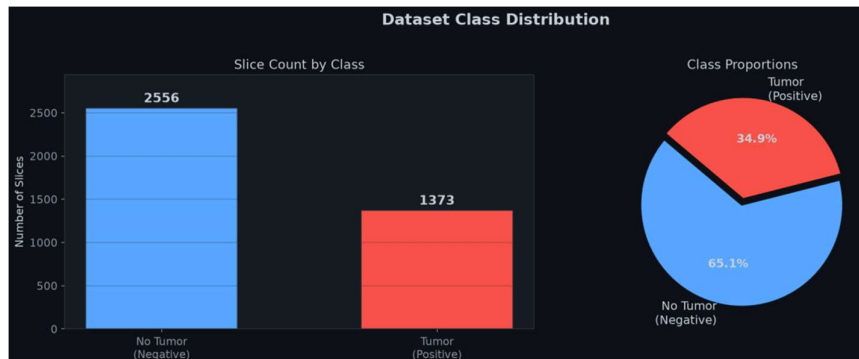
3.1 LGG MRI Segmentation Dataset

The performance of any deep learning model depends heavily on the quality of the dataset used for training and evaluation. In this project, the **LGG MRI Segmentation Dataset** was used. The dataset is publicly available on Kaggle and was developed by **Mateusz Buda et al.** using MRI scans obtained from **The Cancer Imaging Archive (TCIA)**. It contains MRI images of **110 patients** diagnosed with **Lower-Grade Glioma (LGG)** along with manually annotated tumor masks. The dataset primarily consists of **Fluid Attenuated Inversion Recovery (FLAIR)** MRI images, which provide enhanced visualization of abnormal brain tissues by suppressing cerebrospinal fluid signals.

Each MRI slice has a corresponding binary segmentation mask that identifies the tumor region. These masks serve as the ground truth during supervised learning and allow the model to learn the exact location and shape of the tumor.

Table 3.1 Dataset Summary

Parameter	Description
Dataset Name	LGG MRI Segmentation Dataset
Source	Kaggle (Mateusz Buda et al.)
Imaging Modality	MRI (FLAIR)
Tumor Type	Lower-Grade Glioma
Number of Patients	110
Total MRI Slices	Approximately 3,929
Image Format	TIFF (.tif)
Annotation	Manual Binary Segmentation Masks



3.2 Dataset Analysis

Before training the deep learning models, an exploratory analysis of the dataset was performed to understand its structure and quality. The analysis included counting the number of patients, MRI slices, and segmentation masks, as well as verifying that every MRI image had a corresponding ground truth mask. The image dimensions were also inspected to ensure consistency before preprocessing.

Additionally, several MRI images and their corresponding masks were visualized to understand the variation in tumor size, shape, and location across different patients. This analysis helped identify slices without tumors and ensured that the dataset was complete and suitable for model training.

The dataset is organized in a patient-wise directory structure, where each folder contains MRI images and their corresponding segmentation masks.

3.3 Data Preprocessing

Data preprocessing is an essential step before training deep learning models because it improves the consistency and quality of the input images. Several preprocessing operations were performed to prepare the MRI images for training.

Image Resizing

The original MRI images were resized to a fixed resolution before being supplied to the neural networks. Using a uniform image size ensures compatibility with the model architecture and reduces computational requirements during training.

Image Normalization

The pixel intensity values of the MRI images were normalized to the range **0–1** by dividing each pixel value by **255**. Normalization improves numerical stability and enables faster convergence during model training.

Binary Mask Preparation

The segmentation masks were converted into binary images, where:

- **0** represents the background.
- **1** represents the tumor region.

This binary representation simplifies the segmentation problem by allowing the model to distinguish tumor tissue from healthy brain tissue.

Data Augmentation

To improve the robustness of the proposed model and reduce overfitting, several data augmentation techniques were applied during training. These transformations increase the diversity of the training dataset without requiring additional medical images. The augmentation techniques included:

- Horizontal flipping
- Vertical flipping
- Random rotation
- Brightness and contrast adjustment

Dataset Splitting

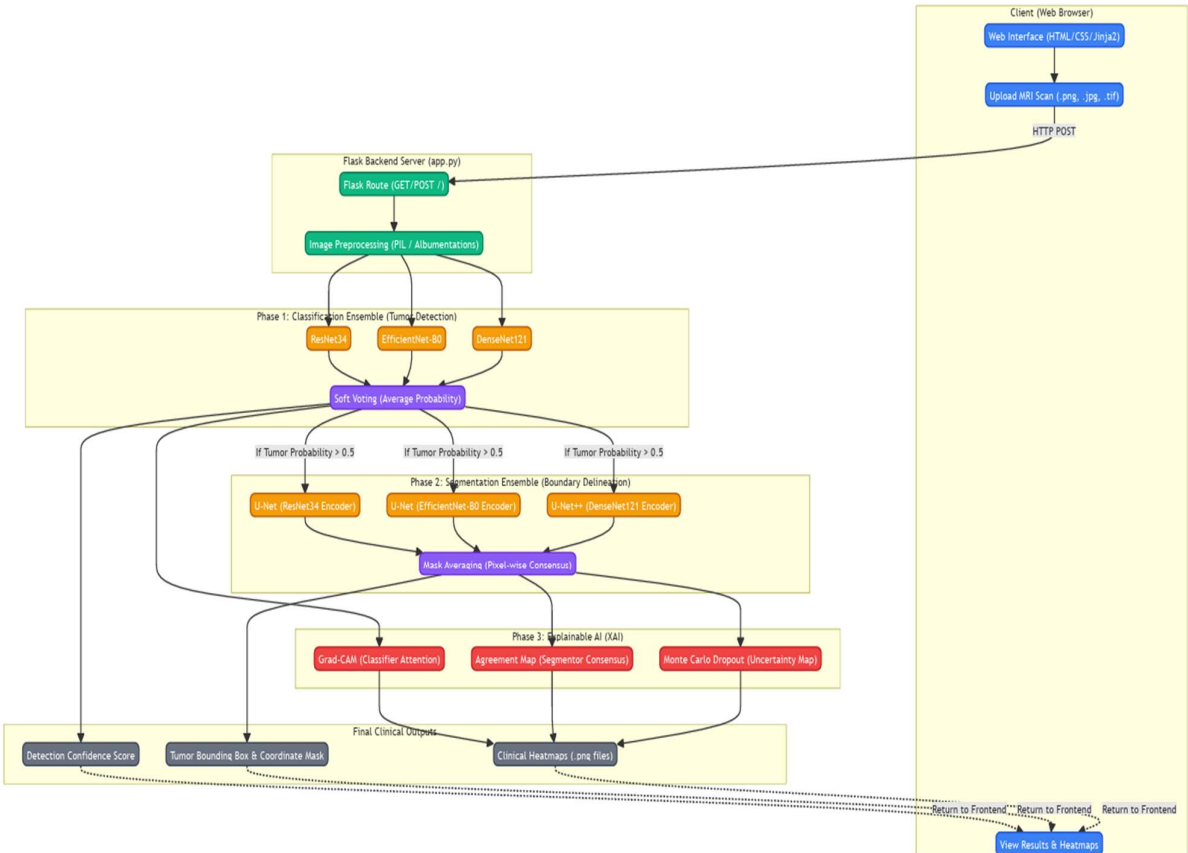
The dataset was divided into **training**, **validation**, and **testing** subsets. A **patient-wise split** was adopted to ensure that MRI slices belonging to the same patient were not present in multiple subsets. This approach prevents data leakage and provides a more realistic evaluation of the model's performance.

Chapter 4

Proposed Methodology

This chapter describes the overall framework developed for automated brain tumor detection and segmentation from MRI images. The proposed system follows a **two-stage pipeline** consisting of a **classification stage** followed by a **segmentation stage**. This approach prevents unnecessary segmentation on healthy MRI slices and improves the reliability of the final predictions. Figure 4.1 illustrates the complete architecture of the proposed system.

Figure 4.1: Overall architecture of the proposed brain MRI segmentation framework.



4.1 Overall Workflow

The proposed framework begins when an MRI image is uploaded through the web interface. The uploaded image is first preprocessed to ensure compatibility with the deep learning models. The preprocessing stage includes image resizing, normalization, and other transformations required before inference.

The preprocessed MRI image is then passed to an ensemble of classification models to determine whether a tumor is present. If the image is classified as healthy, the system returns the prediction directly. If a tumor is detected, the image proceeds to the segmentation stage, where multiple U-Net-based models generate pixel-wise tumor masks. The segmentation results are combined using an ensemble strategy to obtain the final tumor boundary.

Finally, Explainable Artificial Intelligence (XAI) techniques are applied to generate visual explanations, including attention maps and uncertainty estimates. The results are displayed through a Flask-based web application.

4.2 Classification Model

The first stage of the proposed framework focuses on tumor detection. Three pre-trained convolutional neural network models were used:

- ResNet34
- EfficientNet-B0
- DenseNet121

Each model independently predicts the probability of the presence of a brain tumor. During inference, the predictions of all three models are combined using a **soft voting** strategy, where the average probability is calculated. If the final probability exceeds a predefined threshold, the MRI image is classified as tumor-positive and forwarded to the segmentation stage.

Using an ensemble of classifiers improves prediction stability and reduces the possibility of incorrect classifications compared to relying on a single model.

4.3 Segmentation Model

Once a tumor has been detected, the MRI image is processed by the segmentation module to identify the exact tumor region.

The segmentation ensemble consists of:

- U-Net with ResNet34 encoder
- U-Net with EfficientNet-B0 encoder
- U-Net++ with DenseNet121 encoder

Each model generates an independent segmentation mask representing the predicted tumor area. The final segmentation mask is obtained by averaging the outputs of all models, producing a more accurate and robust prediction than any individual network.

This ensemble approach helps reduce segmentation errors and improves boundary detection for tumors with varying shapes and sizes.

4.4 Explainable Artificial Intelligence (XAI)

To improve the transparency of the proposed system, several Explainable Artificial Intelligence (XAI) techniques were incorporated.

Grad-CAM was used to visualize the regions of the MRI image that contributed most to the classification decision.

Agreement Maps were generated to illustrate the consistency between different segmentation models in the ensemble.

Monte Carlo Dropout was employed during inference to estimate prediction uncertainty, allowing the system to identify regions where the model was less confident.

These visual explanations help users understand the model's predictions and increase confidence in the generated results.

4.5 Web Application

A web-based application was developed using the **Flask** framework to provide an easy-to-use interface for the proposed system.

The application allows users to:

- Upload an MRI image.
- Perform automatic tumor detection.
- Generate the predicted segmentation mask.
- View Grad-CAM visualizations and uncertainty maps.
- Display the final results directly in the browser.

The web interface integrates all components of the framework into a single application, making the system convenient for demonstration and future deployment.

Chapter 5

Results and Discussion

5.1 Experimental Setup

The proposed framework was implemented using **Python** and the **PyTorch** deep learning library. Model training and evaluation were carried out on a workstation equipped with an **NVIDIA GeForce RTX 3050 Laptop GPU (4 GB VRAM)** with CUDA acceleration. The Adam optimizer was used for training with an initial learning rate of **0.0001**. To improve convergence and reduce overfitting, a **ReduceLROnPlateau** learning rate scheduler and an **Early Stopping** strategy were employed during training.

The classification models were trained using **Binary Cross Entropy (BCE) Loss**, while the segmentation models were optimized using a combination of **Dice Loss** and **Binary Cross Entropy (BCE) Loss** to achieve accurate pixel-wise segmentation and improved overlap with the ground truth masks

5.2 Performance Evaluation

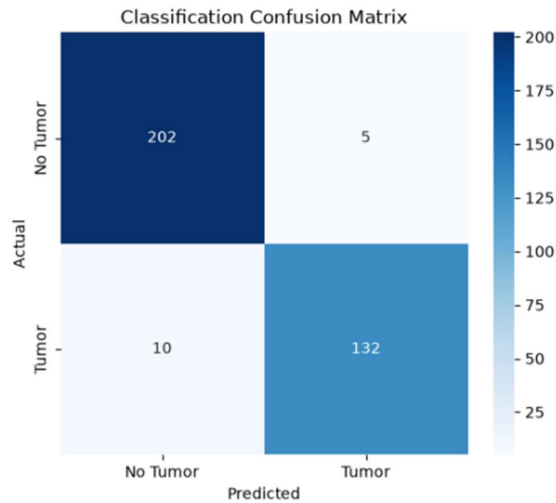
The proposed system was evaluated on a separate test set containing MRI images that were not used during training.

Classification Performance

The ensemble classification model achieved high performance in distinguishing MRI images containing tumors from healthy images.

Table 5.1 Classification Performance

Metric	Value
Accuracy	95.70%
Precision	96.15%
Recall	92.59%
F1-Score	94.62%
ROC-AUC	98.15%



Segmentation Performance

The segmentation ensemble successfully identified tumor boundaries with a high degree of overlap with the manually annotated ground truth.

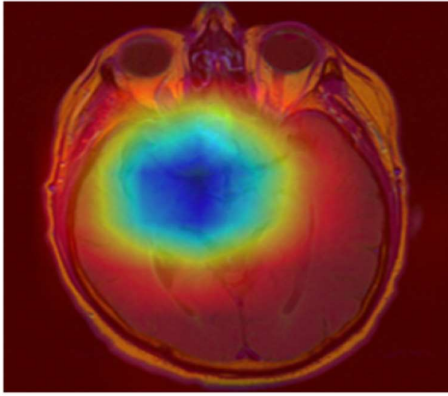
Table 5.2 Segmentation Performance

Metric	Value
Dice Score	88.33%
Intersection over Union (IoU)	79.53%

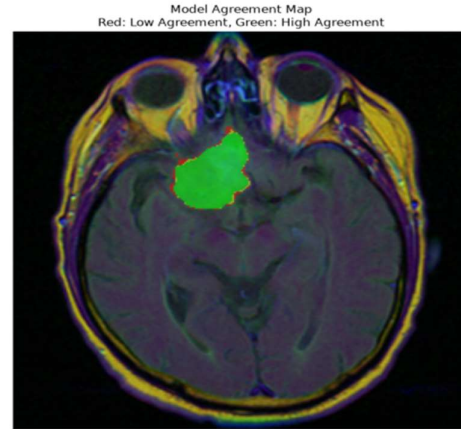
The obtained Dice Score indicates that the proposed ensemble model can accurately segment tumor regions while maintaining good agreement with expert annotations.

5.3 Sample Results

Following Figure presents representative outputs produced by the proposed framework. The system correctly detects the presence of a tumor, generates the corresponding segmentation mask, and provides visual explanations through Explainable Artificial Intelligence (XAI) techniques.



(a) Grad-CAM highlighting classifier focus.



(b) Segmentor Agreement Map.

5.4 Discussion

The experimental results demonstrate that combining multiple deep learning models improves both classification and segmentation performance compared to relying on a single model. The use of an ensemble strategy increases the robustness of predictions and reduces false detections.

Furthermore, the integration of Explainable Artificial Intelligence techniques, such as **Grad-CAM**, **agreement maps**, and **Monte Carlo Dropout**, provides visual explanations that improve the interpretability of the model's predictions. These explanations can assist clinicians in understanding and validating the automated results.

Overall, the proposed framework achieved reliable performance while remaining suitable for deployment on consumer-grade hardware, demonstrating its potential as a decision-support tool for brain tumor analysis.

Chapter 6

Conclusion and Future Scope

Conclusion

This internship project presented a deep learning-based framework for the automatic detection and segmentation of **Lower-Grade Glioma (LGG)** from brain MRI images. The proposed system combines an ensemble of classification models with U-Net-based segmentation models to accurately identify tumor regions while reducing false predictions.

The framework achieved a **classification accuracy of 95.70%** and a **Dice Score of 88.33%** on the test dataset, demonstrating its effectiveness for automated brain tumor analysis. The integration of Explainable Artificial Intelligence techniques, including **Grad-CAM**, **agreement maps**, and **Monte Carlo Dropout**, provided visual explanations that improved the transparency of the model's predictions.

Finally, a **Flask-based web application** was developed to integrate the complete workflow, allowing users to upload MRI images, perform automated analysis, and visualize the results through an interactive interface. The outcomes of this project demonstrate the potential of deep learning as a reliable decision-support tool for medical image analysis and provide a strong foundation for future research and development

6.1 Challenges Faced

During the development of this project, several challenges were encountered. The primary challenge was the limited computational resources available for training deep learning models. The entire framework was developed and trained on a system equipped with an **NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB VRAM**, which imposed strict memory constraints. To overcome this limitation, the models were trained sequentially instead of simultaneously, and **Automatic Mixed Precision (AMP)** was used to reduce GPU memory usage and improve training efficiency.

Another challenge was the limited size of the LGG MRI Segmentation dataset. Since medical imaging datasets are relatively small compared to general computer vision datasets, there was a risk of overfitting. This issue was addressed through appropriate data preprocessing, data augmentation, and careful selection of training and validation samples.

6.2 Future Scope

The proposed framework can be further improved in several ways.

- Extend the system to support **3D MRI volumes**, allowing the model to utilize spatial information across consecutive MRI slices.
 - Incorporate additional MRI modalities such as **T1**, **T1ce**, and **T2** along with **FLAIR** images to improve segmentation accuracy.
 - Train and evaluate the framework on larger and more diverse medical imaging datasets to improve generalization.
 - Deploy the application on a cloud platform to enable remote access and real-time clinical use.
 - Investigate more advanced deep learning architectures and Explainable AI techniques to further enhance prediction accuracy and model interpretability.
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